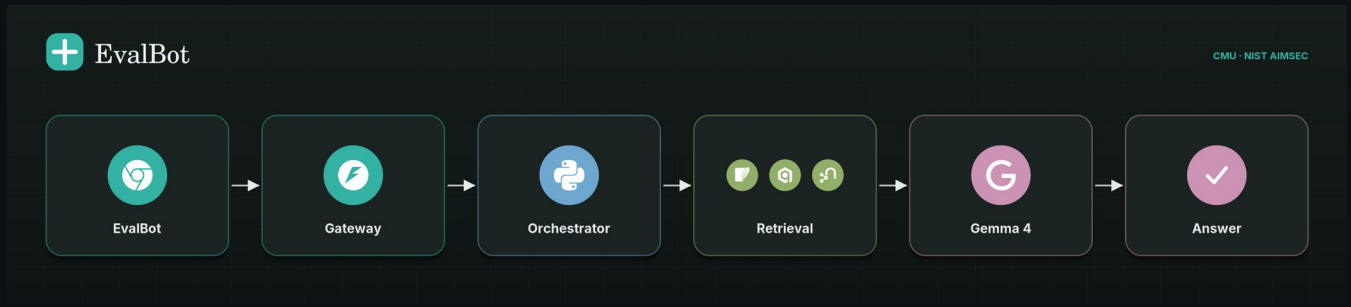


An evaluation assistant for clinical AI

Ask what to evaluate for a clinical model. It answers with a cited evaluation plan, a checklist you can tick off, and the standard that applies.

CMU · NIST AIMSEC | **Gemma 4 26B-A4B** | **15,954 papers**

github.com/adi274903/MARS-Healthcare · live at localhost:8080

THE GAP

Evaluation is scattered. Claims go out unaudited.

— **Reporting guidance is splintered.** TRIPOD+AI, PROBAST+AI, CONSORT-AI, DECIDE-AI, CLAIM, CHART.

— **Governance sits elsewhere.** NIST AI RMF, FUTURE-AI, WHO, FDA GMLP/PCCP/SaMD, CHAI, EU AI Act, ISO/IEC 42001.

— **The evidence of what the field actually did** is buried in 15,954 papers nobody has time to read.

— **So every team invents its own checklist,** and a model card becomes an act of memory rather than a citation.

the problem is retrieval + synthesis, not a lack of knowledge

WHAT IT DOES

Four things, on every question

01

Plans

Turns “should we trust this model?” into concrete, checkable evaluation items.

02

Cites

Every item points at a PubMed or DOI record, with the claim it supports attached.

03

Names the standard

TRIPOD+AI, PROBAST+AI, FDA SaMD, EU AI Act, NIST AI RMF — matched, not guessed.

04

Counts exactly

Corpus totals come from SQL, never from the model's arithmetic.

HOW ONE QUESTION FLOWS

Routed by intent before a single token is generated



WHY THE ROUTER MATTERS

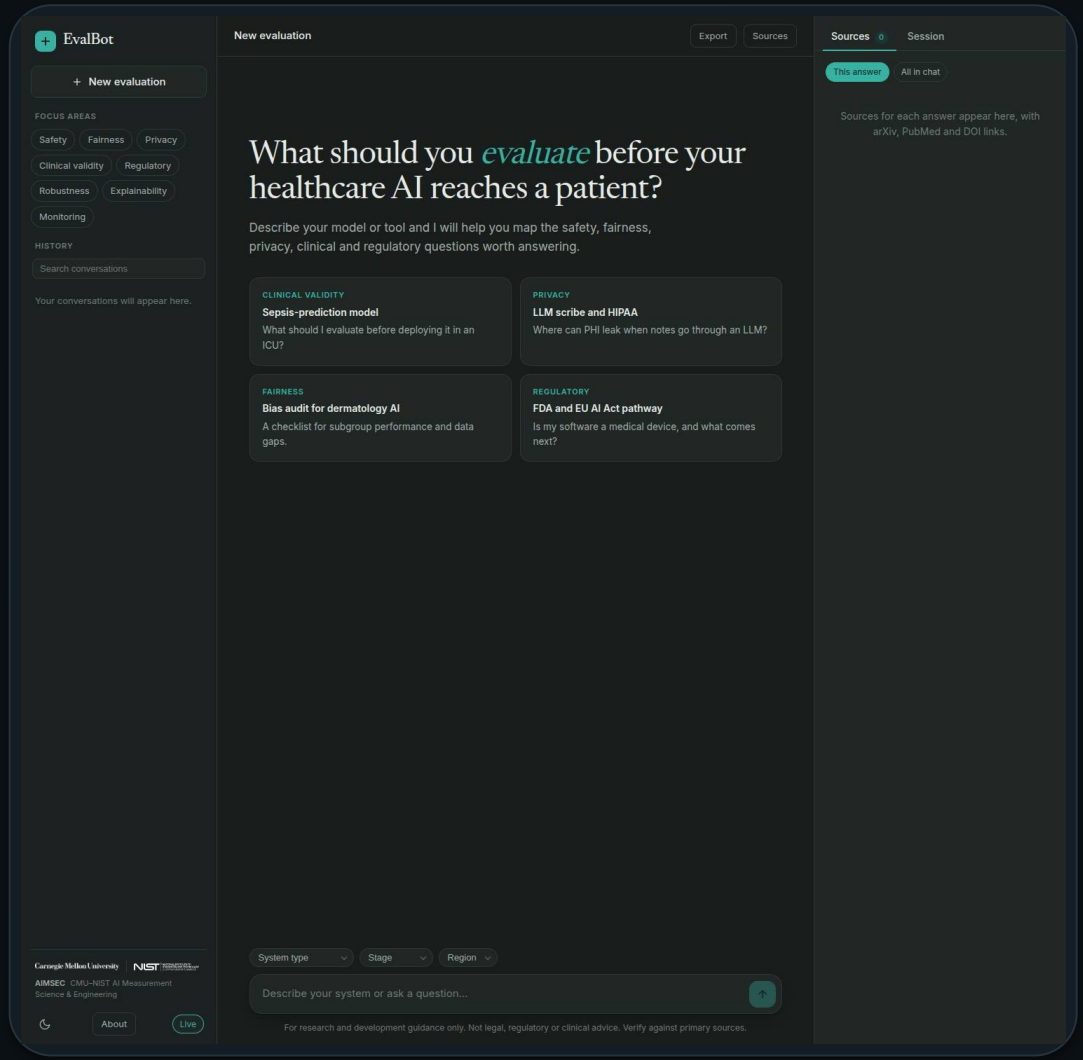
— **“How many papers…” never reaches the model.** It is a SQL aggregate, quoted verbatim.

— **A question about a regulation** pulls guidance, not an arbitrary paper.

DEMO · 1

Start from a clinical system

Pick system type, stage and region; ask in plain language. The UI shows how many papers the answer stands on.

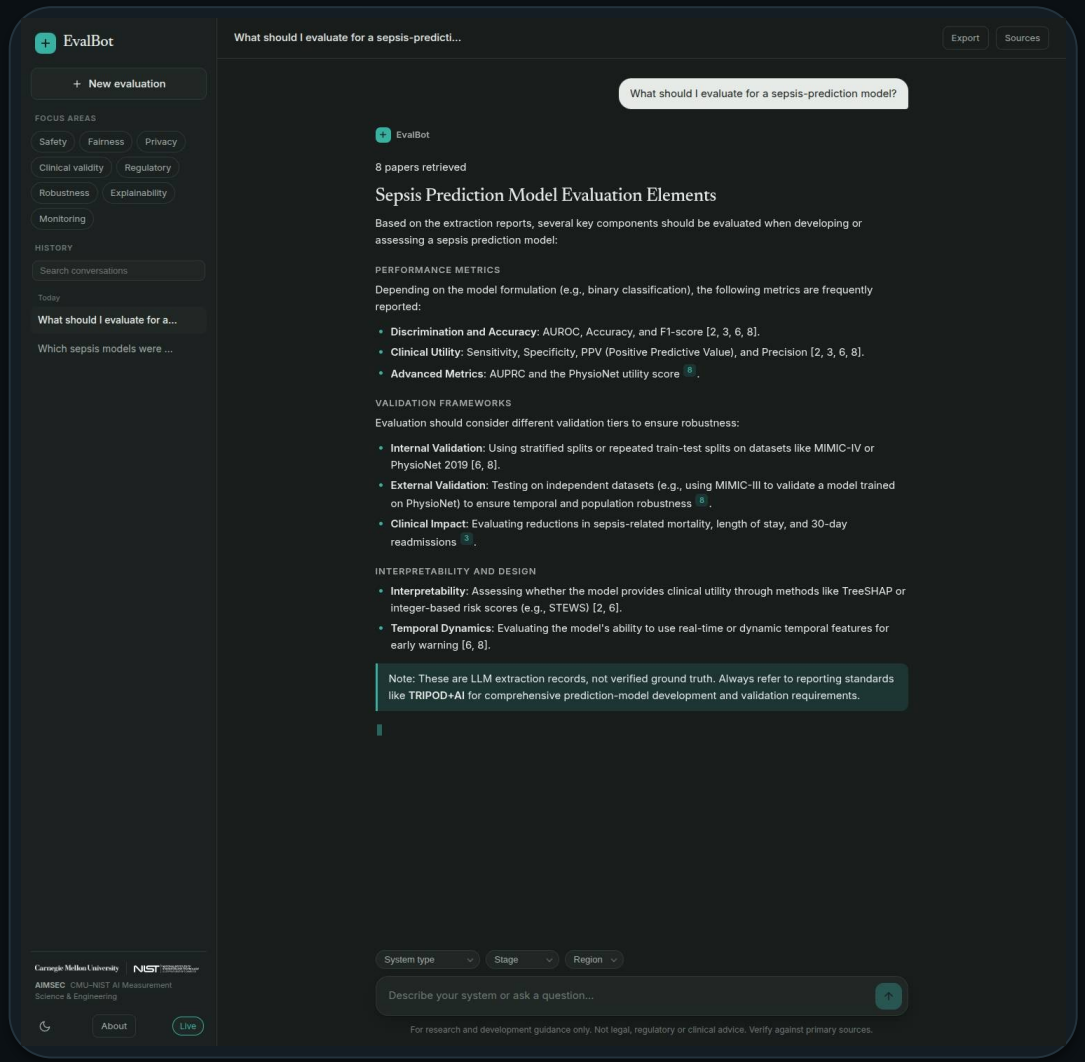


live capture from localhost:8080 · Gemma 4 26B-A4B via OpenRouter

DEMO · 2

It plans the evaluation, live

Performance metrics, validation frameworks, interpretability — generated as you watch, with citations appearing inline.

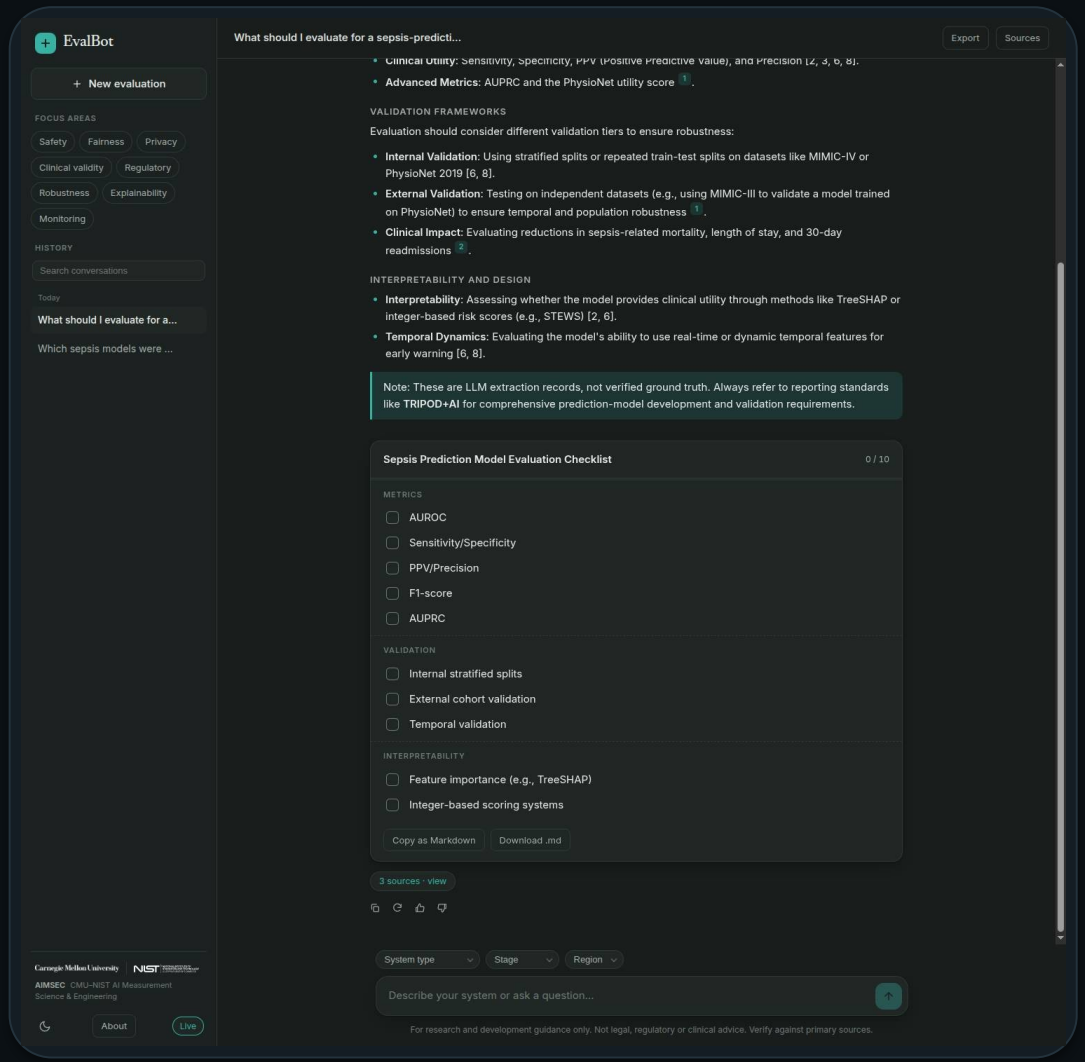


live capture from localhost:8080 · Gemma 4 26B-A4B via OpenRouter

DEMO · 3

The deliverable: a checklist

10 items across metrics, validation and interpretability.
Tick them off, copy as Markdown, or download the file.

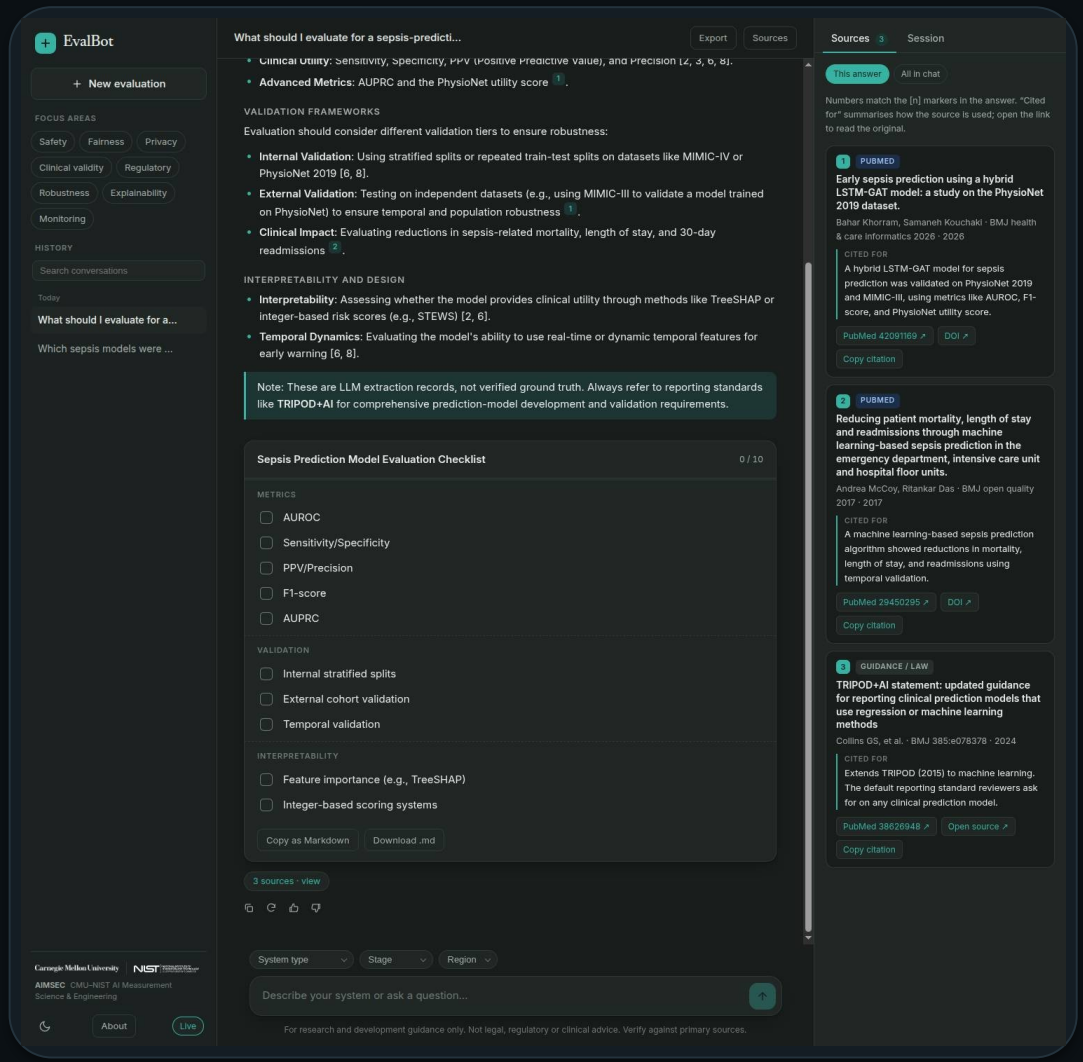


live capture from localhost:8080 · Gemma 4 26B-A4B via OpenRouter

DEMO · 4

Every item is traceable

PubMed records with PMID and DOI, and a GUIDANCE / LAW card for the standard that applies — here TRIPOD+AI.

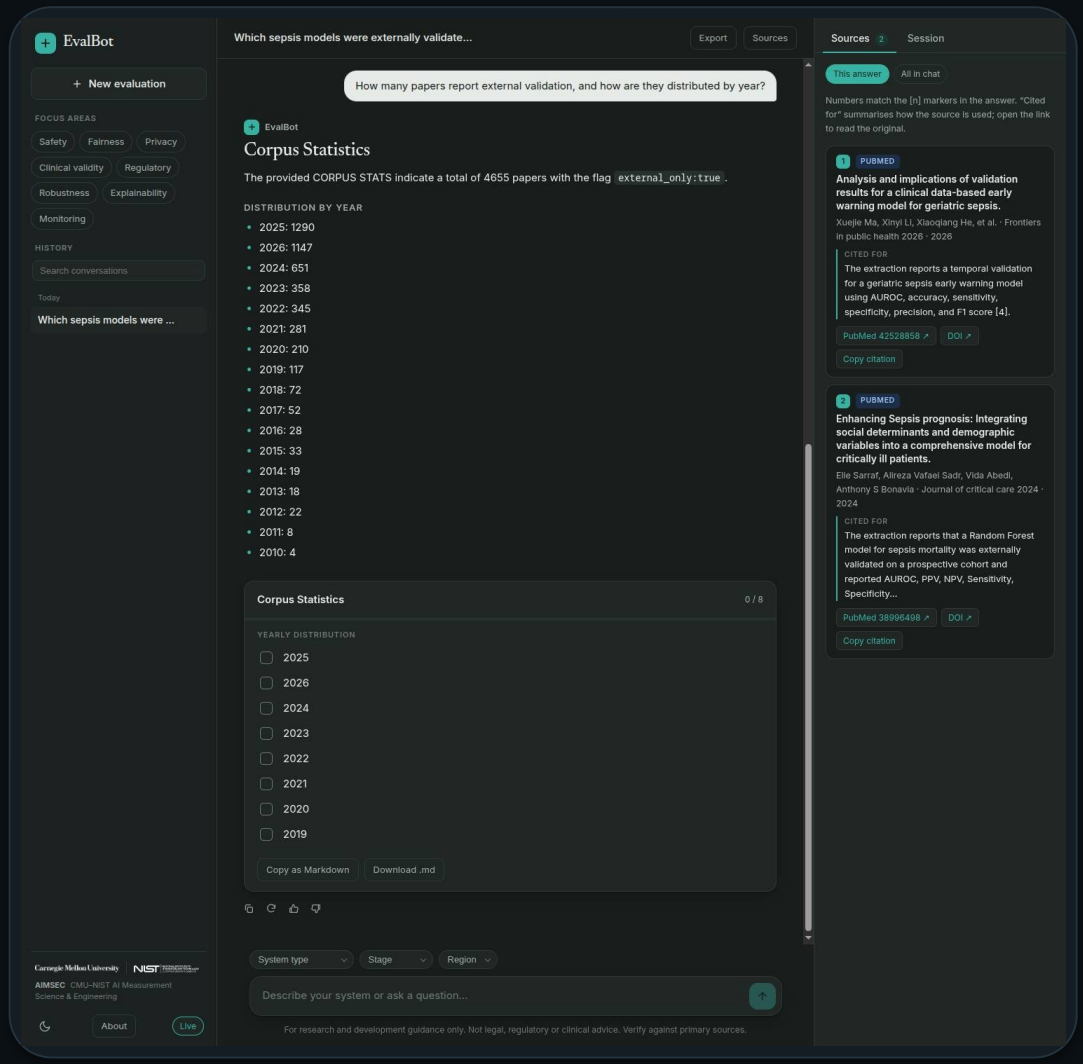


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DEMO · 5

Counts come from SQL

“How many papers report external validation?” → 4,655, with a year breakdown. Not a guess.



live capture from localhost:8080 · Gemma 4 26B-A4B via OpenRouter

WHAT IS INSIDE

Numbers a reviewer can check

15,954
papers
7 extraction axes

4,655
externally validated
flag, not text match

24
frameworks
8 reporting · 6 governance

124
offline tests
no network in CI

— **Cross-store consistency:** 15,954 rows in JSONL and SQLite, 15,954 papers in the graph (100,185 authors, 6,068 MeSH terms).

— **Flags reproduce exactly:** external 4,655 · temporal 1,352 · internal 14,738 · deployed 628 · RCT 368 · CI 8,348.

numbers printed by the build, not remembered

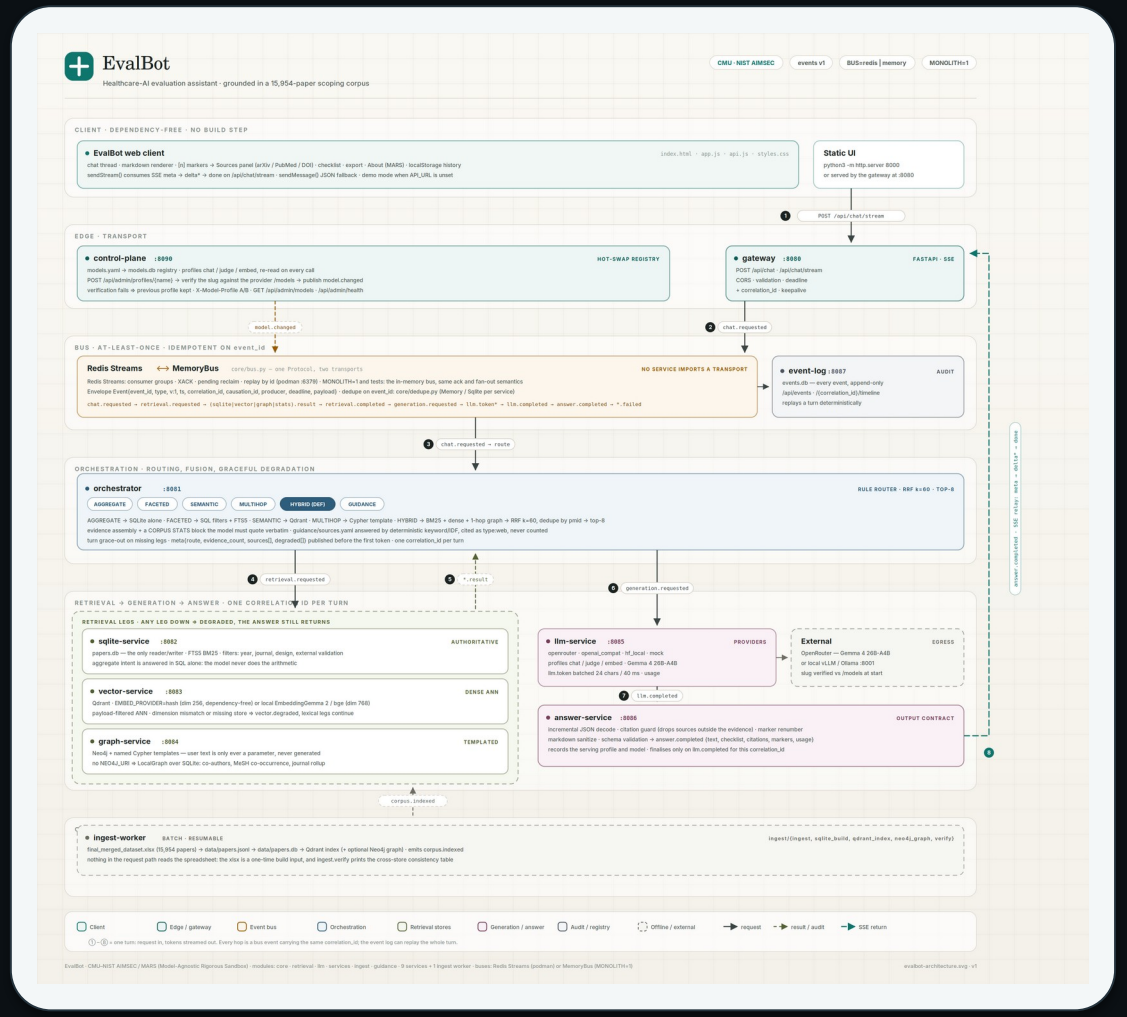
HOW IT IS BUILT

Nine services, one event bus

— **SQLite** is the system of record: 15,954 rows, FTS5, every exact count.

— **Qdrant** holds 768-d EmbeddingGemma-2 vectors; Neo4j holds the paper graph.

— **Both are derived:** rebuildable from the JSONL, and the answer still works if either is down.



same code runs as one process (MONOLITH=1) or nine over Redis Streams

THE PARTS THAT MATTER

Three stores, one rule: the record is the source of truth

SQLite

SYSTEM OF RECORD

15,954 rows · FTS5 BM25 · every exact count and facet filter. Read-only at request time.

Qdrant

DENSE RECALL

768-d EmbeddingGemma-2 with asymmetric query/document prompts; bf16 on an 8 GB GPU.

Neo4j

MULTI-HOP GRAPH

papers, authors, journals, MeSH. Named Cypher templates only — with a SQLite fallback.

Qdrant and Neo4j are derived and rebuildable — SQLite stays authoritative

THE SAFETY BEHAVIOUR

It refuses more than a chatbot does

— **No invented citations.** Citations are built from retrieved records; markers outside the evidence are dropped.

— **No invented numbers.** Corpus statistics are SQL results quoted verbatim.

— **No false authority.** Extraction fields are labelled unverified — 40-paper audit found CI fields wrong ~70% of the time.

— **No silent failure.** A missing leg is reported as degraded; a missing model returns a readable error.

cite-or-refuse is the product, not a feature flag

What should we evaluate, and what does the evidence say?

Ask it about your own model. Every answer is cited or refused; every number comes from the corpus; the model behind it is swappable at runtime.

github.com/adi274903/MARS-Healthcare

demo video + PDF